

Lab for individually and block randomized control trials

Chapters 7 and 8

Case Study A

Museum Science Spatial: A Program for Pre-Schoolers

A STEM Museum in Florida has had a longstanding partnership with two researchers at a nearby university. Drs. Andrews and Zheng are interested in how informal experiences in museums may encourage families to engage with their children in ways that inspire and support STEM learning at home. This interest has led them to work with the museum staff to design a series of exhibits that foster interactions (while at the museum), while also providing materials to take home, with the hope of continued engagement. Understanding if this program actually improves these connections is important, given the continued cost of these take-home materials. This study focuses on a particular exhibit that focuses on fostering the development of spatial skills through a series of building activities.

Plausible Evaluation Design A

Individual Random Assignment

Suppose researchers wish to evaluate the effectiveness of a similar program on parent interaction with their children at home, measured using a well-established validated scale based on questions answered through an online survey collection system. The design is to recruit families as they enter the museum into the study, then randomly assign them to attend the additional exhibits and receive the additional take home materials (treatment group) or attend the typical exhibits (control group). After 30 days, all study families take a survey and their answers are used to compute an outcome score. These scores are then analyzed using an independent samples t test to compare the averages of treatment and control families.

For simplicity, assume perfect compliance with random assignment and a perfect response rate.

Question A-1

Using the planner tool, find the minimum detectable effect for a study in which 20 families are assigned to the treatment group and 20 families are assigned the control group when power is .8 for a test with a significance level of .05.

```
compute_mdes(V = srs_fixed_nocovariate_V(20,20),
             df=srs_fixed_nocovariate_df(20,20),
             beta = 1-.8,
             alpha = .05
)
```

```
## [1] 0.9089294
```

Question A-2

A member of the study team claims that a better study would use a higher portion of the study families as treatment. Using the planner tool, find the minimum detectable effect for a study in which 30 families are

assigned to the treatment group and 10 families are assigned the control group when power is .8 for a test with a significance level of .05. Is this design more sensitive?

```
compute_mdes(V = srs_fixed_nocovariate_V(30,10),
             df=srs_fixed_nocovariate_df(30,10),
             beta = 1-.8,
             alpha = .05
)
```

```
## [1] 1.049541
```

Question A-3

There is some concern that the study may not be as sensitive as the researchers need it to be. The **first idea** is to use 60 families, with 30 in treatment and 30 in control, without a covariate. The **second idea** is to have each of the 40 study members fill out a small questionnaire that would lead to a covariate correlated with the outcome at $\rho = 0.5$ ($\rho^2 = 0.25$).

Using the planner tool, find the minimum detectable effect for a study in which 30 families are assigned to the treatment group and 30 families are assigned the control group, without using a covariate, when power is .8 for a test with a significance level of .05. This is the sensitivity of the **first idea**.

```
compute_mdes(V = srs_fixed_nocovariate_V(nt = 30,nc = 30),
             df=srs_fixed_nocovariate_df(nt = 30,nc = 30),
             beta = 1-.8,
             alpha = .05
)
```

```
## [1] 0.7354893
```

Using the planner tool, again, find the minimum detectable effect for a study in which 20 families are assigned to the treatment group and 20 families are assigned the control group, with a covariate is employed where ($\rho^2 = 0.25$), when power is .8 for a test with a significance level of .05. This is the sensitivity of the **second idea**.

```
compute_mdes(V = srs_fixed_covariate_V(nt = 20, nc = 20, R2_1 = .25, q = 1),
             df=srs_fixed_covariate_df(nt = 20, nc = 20, q = 1),
             beta = 1-.8,
             alpha = .05
)
```

```
## [1] 0.7877171
```

Which idea is more sensitive?

Case Study B

High School After-School Robotics Program

Dr. Carlos – a computer scientist – and his students have been running an after-school outreach program focused on robotics at a nearby high school in Atlanta. This high school predominately serves Black students – which are underrepresented in computer science – and the students come from both middle-class and low-SES families. In order to be admitted to the program, which is free, students had to apply and be recommended by a teacher. Once admitted, students in the program meet once a week for 2 hours for 16 weeks. During the lessons, they learn to program and build robots to complete tasks and, at the end of the program, they take part in a mini ‘competition’. The research team has been excited to see how the students have responded to the program, including how this has affected their interest in computer science in general, as well as in

studying a STEM related field in college. Teachers have reported that students in the program are more engaged in their science classes.

Plausible Evaluation Design B

Random Assignment Within Blocks (Schools): Fixed Effects

Suppose researchers wish to evaluate the effectiveness of a similar program on student science grades in school. Researchers enroll students in the study during the the fall term. Students who are enrolled into the study are randomized within each school into either the treatment condition (the program) or a control condition (whatever they planned to do after school). Spring science scores on a state-mandated standardized test are used as the outcome. A set of schools are recruited for the study from a variety of districts through a professional development conference for state principals and district membership is not considered in the design. For planning purposes assume block effects explain about 10 percent of the total variation.

For simplicity, assume perfect compliance with random assignment and a perfect response rate.

Question B-1

Using the planner tool, find the minimum detectable effect for a randomized block study with fixed effects that employed 12 schools, in each of which 10 students are assigned to the treatment group and 30 students are assigned the control group when power is .8 for a test with a significance level of .05.

```
compute_mdes(V = dummyblocked_fixed_nocovariate_V(m = 12,
                                                    nt = 10,
                                                    nc = 30,
                                                    icc = .10),
            df = dummyblocked_fixed_nocovariate_df(m = 12,
                                                    nt = 10,
                                                    nc = 20),
            beta = 1-.8,
            alpha = .05
            )
```

```
## [1] 0.2809016
```

Question B-2

A member of the study team claims that a better study would use a similar portion of students in treatment and control condition. Using the planner tool, find the minimum detectable effect for a randomized block study with fixed effects that employed 12 schools, in each of which 20 students are assigned to the treatment group and 20 students are assigned the control group when power is .8 for a test with a significance level of .05. Is this design more sensitive?

```
compute_mdes(V = dummyblocked_fixed_nocovariate_V(m = 12,
                                                    nt = 20,
                                                    nc = 20,
                                                    icc = .10),
            df = dummyblocked_fixed_nocovariate_df(m = 12,
                                                    nt = 15,
                                                    nc = 15),
            beta = 1-.8,
            alpha = .05
            )
```

```
## [1] 0.2432679
```

Question B-3

After signing up the schools, it was apparent that using a 10th grade math test as a covariate was possible for the analysis. Earlier analyses of other cohorts show that the 10th grade math test was correlated with the focal science test at $\rho = .8$ for an R-square value of $\rho^2 = .64$. The hope is that having this covariate can reduce the total number of schools needed from 12 to 10.

Using the planner tool, find the minimum detectable effect for a randomized block study with fixed effects that employed 10 schools, in each of which 20 students are assigned to the treatment group and 20 students are assigned the control group for an analysis that also includes the earlier math score, when power is .8 for a test with a significance level of .05. Is this design more sensitive?

```
compute_mdes(V = dummyblocked_fixed_covariate_V(m = 10,
                                                    nt = 20,
                                                    nc = 20,
                                                    icc = .10,
                                                    R2_1 = .64,
                                                    q = 1),
              df = dummyblocked_fixed_covariate_df(m = 10,
                                                    nt = 15,
                                                    nc = 15,
                                                    q = 1),
              beta = 1-.8,
              alpha = .05
            )
```

```
## [1] 0.1599794
```

References

- Barker, B. S., & Ansorge, J. (2007). Robotics as means to increase achievement scores in an informal learning environment. *Journal of research on technology in education*, 39(3), 229-243.
- Marcus, M., Haden, C.A., & Uttal, D.H. (2017) STEM Learning and transfer in a children's museum and beyond. *Merrill-Palmer Quarterly*, 63(2): 155-180.